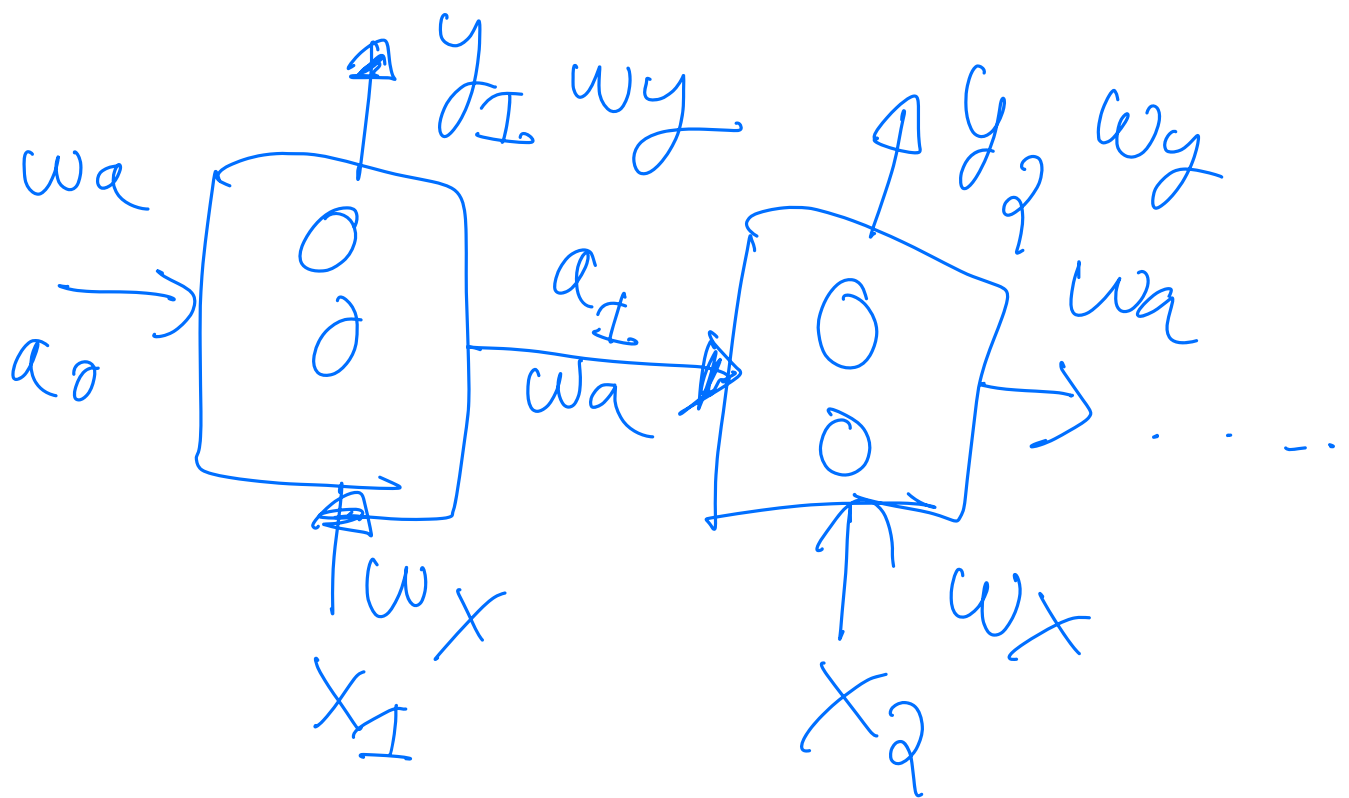
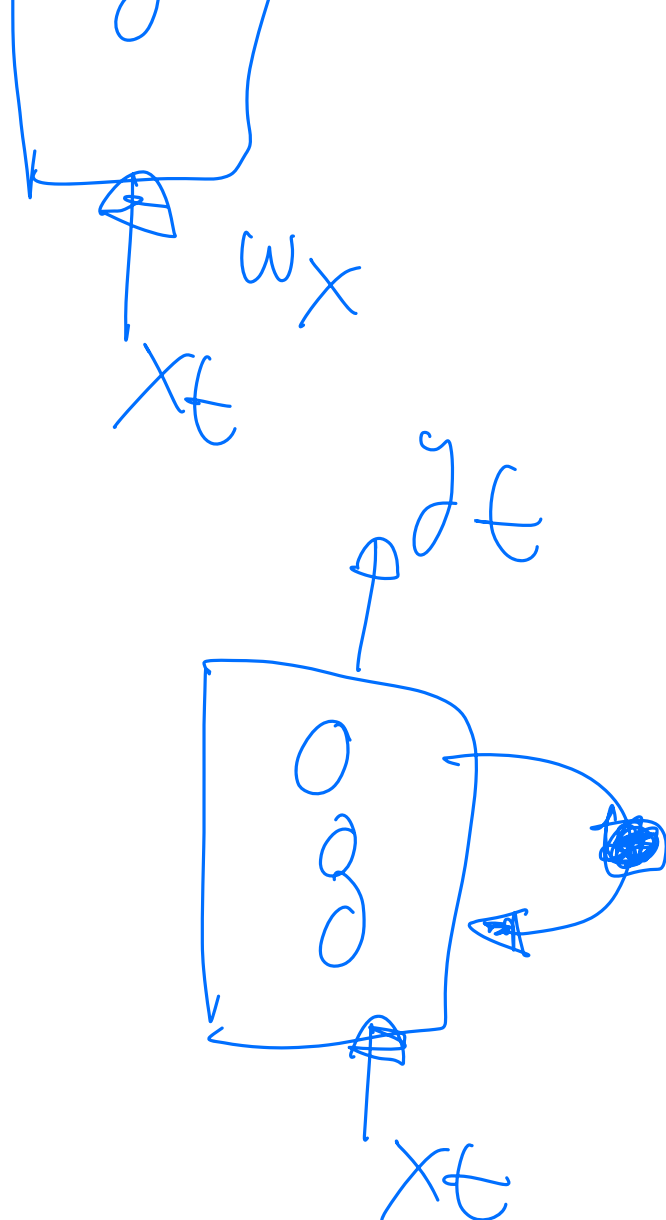


RNN

↓ / Basic RNN Cell



$\Rightarrow$



$$a_t = g(w_a a_0 + w_x x_t + b_a)$$

$$g_t = g'(w_g a_t + b_g)$$

$$g = \{ \tanh \mid \text{ReLU} \}$$

$$g' = \text{Sigmoid} \mid \text{softmax}$$

$\Rightarrow$

$$a_t = g \left(W a_{t-1}, x_t \right) + b_a$$

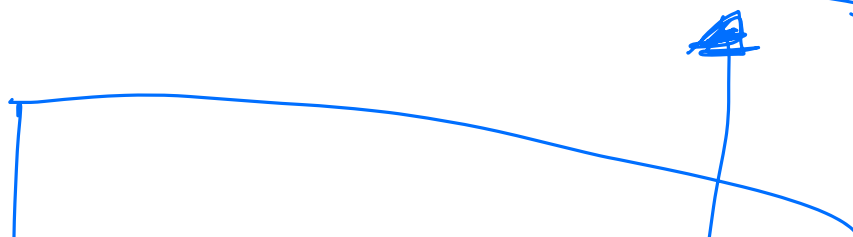
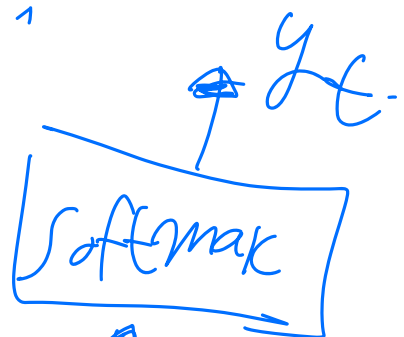
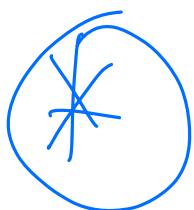
$$Loss_t(g_t^{\hat{}} / g_t) =$$

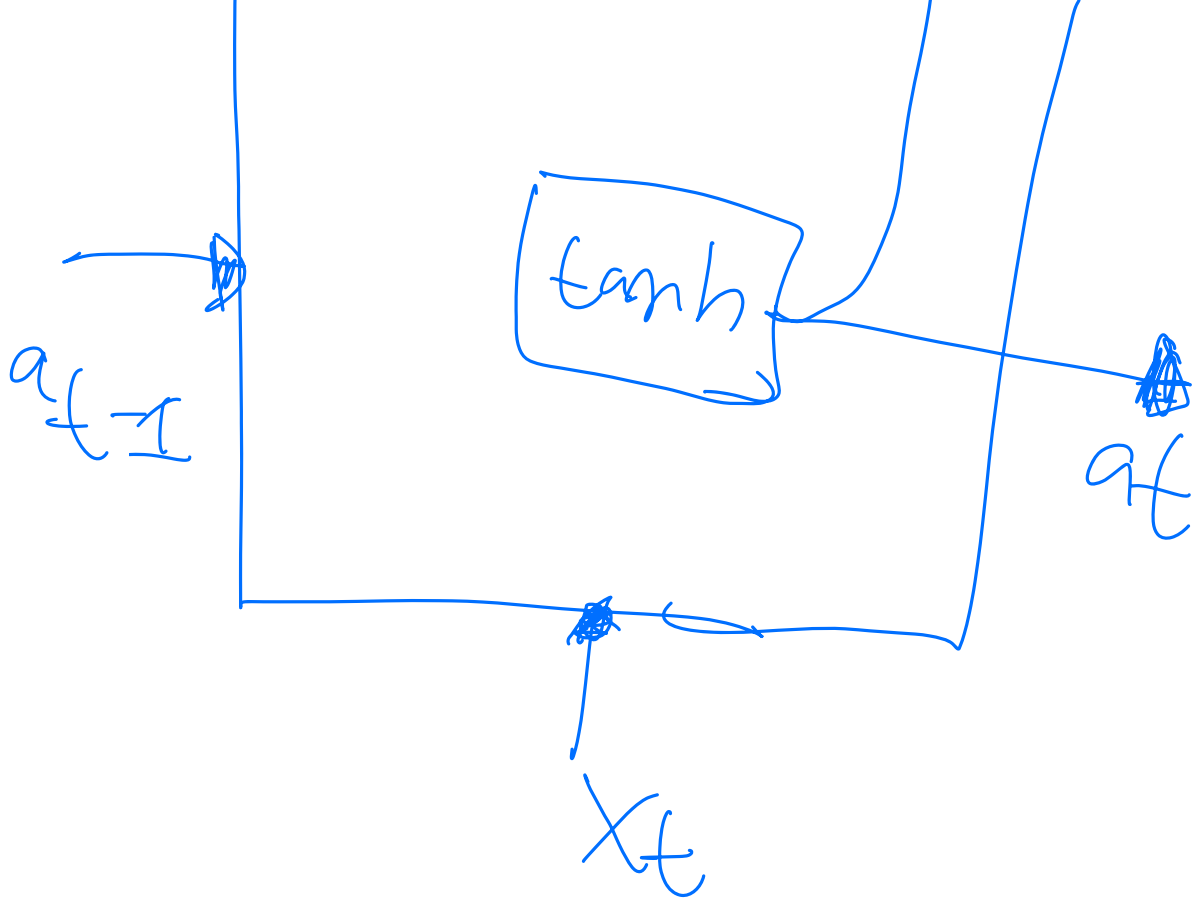
$$-y_t \log y_t^1 - (1-y_t) \cdot$$

$$\log(1-y_t^1)$$

$$\text{Net loss} = \sum_{t=1}^T \text{loss}_t$$

* Vanishing gradients





$$a_t = \tanh \left(w a_{t-1} + b a \right)$$

GRU

z = memory cell

(additional parameter)

$$z_t = a_t$$

$$\textcircled{1} \quad \tilde{z}_{(\text{candidate})} = \tanh(w_c(z_{t-1}, x_t) + b_c)$$

$$\textcircled{2} \quad \Gamma_u = \sigma(w_u(z_{t-1}, x_t))$$

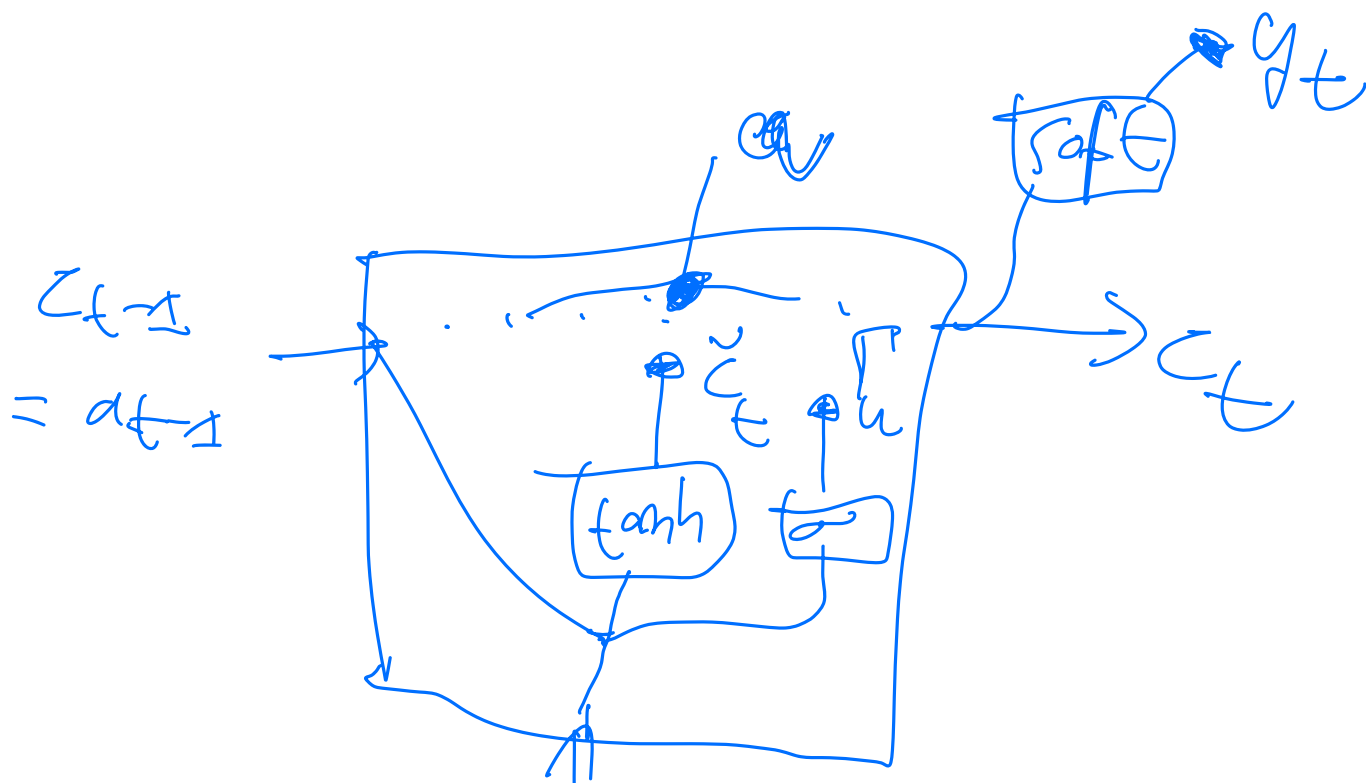
(gate)
update
 $0 \leq \Gamma_u \leq 1$

$+ b_u$

③

$$c_t = \Gamma_u \times \tilde{c}_t + (1 - \Gamma_u) \cdot c_{t-1}$$

GRU equation



x_t

$\Rightarrow$

$$z_t = \tanh \left(w_c \left[\vec{r} * c_{t-1} \right] + b_z \right)$$

$$\vec{r} = \sigma \left(w_r (c_{t-1}, x_t) + b_r \right)$$

LSTM



$$\textcircled{1} \quad \tilde{c}_t = \tanh(w_c[a_{t-1}, x_t] + b_c)$$

$$\textcircled{2} \quad \tilde{u} = \sigma(w_u[a_{t-1}, x_t] + b_u)$$

$$\textcircled{3} \quad \tilde{f} = \sigma(w_f[a_{t-1}, x_t] + b_f)$$

$$\textcircled{4} \quad r_o = \sigma(w_o(a_{t-1}, x_t) + b_o)$$

$$\textcircled{5} \quad z_t = p_u * c_t + p_f * c_{t-1}$$

$$\textcircled{6} \quad a_t = p_o * z_t$$

Bi directional RNN

forward and

backward

activation